

Question number	Answer	Additional guidance	Mark
9(a)	An explanation that includes three of the following points: - because {it / light} is needed for photolysis (1) - because {electrons released (from the water) are needed to replace those lost from the photosystems / hydrogen ions are used to reduce NADP} (1) - because {it / light} is needed to {excite / release} the electrons of the {photosystems / chlorophyll / PS / photosynthetic pigments} (1) - electrons (from photosystems) used to produce ATP (and reduced NADP) (1)	ACCEPT description of photolysis ACCEPT protons / H⁺ to produce {reduced NADP / NADPH}	(3)

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9(b)(i)	The correct answer is D (sunlight) <i>A is incorrect because green light would have been a negative control</i> <i>B is incorrect because green light is not included in the study</i> <i>C is incorrect because ultra violet is not part of the visible spectrum</i>		(1)

Question number	Answer	Additional guidance	Mark
9(b)(ii)	An explanation that includes four of the following points: • high rate of photosynthesis in {red / blue and red} light (1) • because there is more (light) energy available (for photosynthesis) (1) • leaf biomass will be high when photosynthesis is fast (1) • as more {glucose / GALP} synthesised to produce {(insoluble) organic molecules / biomass} (1) • the control light does not have as much red and blue light so {the rate of photosynthesis will not be so high / leaf biomass will be lower} (1)	ACCEPT low rate of photosynthesis in amber light ACCEPT converse for amber ACCEPT converse ACCEPT converse	(4)

Question number	Answer	Additional guidance	Mark
9(b)(iii)	An answer that includes three of the following points: - all three types of pigment are found regardless of the type of light shone on the plants (1) - the type of light shone at the plants affects the concentrations of pigments found (1) {chlorophyll a is the most abundant / anthocyanins are the least abundant} (in all four groups / in total) (1) - red light results in the highest (total) concentration of pigments / amber light results in the lowest concentration of pigments (1) - anthocyanin content least {affected / varied} <u>by the type of light</u> (1)	ACCEPT mass for concentration throughout ACCEPT type of light affects each pigment differently ACCEPT proportion / amount / highest / lowest ACCEPT all had similar concentrations	(3)

Question number	Answer	Additional guidance	Mark
9(b)(iv)	An explanation that includes the following points: • credit two appropriate genes named (1) • credit correct explanation for one named gene (1) • credit correct explanation for the other named gene (1)	e.gs ATP synthase : phosphorylation of ADP chlorophyll (a) (production) : absorption of light / to release electrons / table shows increased chlorophyll a concentrations carotene (production) : absorption of light / to release electrons RUBISCO: carbon fixation / description ETC (proteins) : pump protons into thylakoid (space) / pass protons to NADP to reduce it RNA polymerase: to synthesise RNA enzymes to catalyse reduction of NADP : so reduced NADPH can be produced for conversion of GP into GALP named enzyme in Calvin Cycle : reaction specified NB Explanations for chlorophyll and carotene must be different to award full marks	(3)

